

```
(%i6) /* Initialization */
declare(h, real)$;
assume(h > 0)$;
declare([C_1, C_2], [constant, real])$;
declare(p, [constant, integer])$;
assume(p > 0)$;
depends(D, h)$;
```

```
(%i9) /* Modeling true derivative */
q1: D(h) + C_1 · h^p + C_2 / h;
q2: subst(h = h / 2, q1);
q4: subst(h = h / 4, q1);
```

q1 $D(h) + C_1 h^p + \frac{C_2}{h}$

q2 $\frac{C_1 h^p}{2^p} + \frac{2 C_2}{h} + D\left(\frac{h}{2}\right)$

q4 $\frac{C_1 h^p}{4^p} + \frac{4 C_2}{h} + D\left(\frac{h}{4}\right)$

```
(%i10) /* q1, q2 and q3 must be the same thing. Compute C1 and C2 */
C1C2: solve([q1 = q2, q2 = q4], [C_1, C_2])[1];
```

C1C2
$$\left[C_1 = - \frac{\left(2^{p+1} 4^p D(h) - 3 2^p 4^p D\left(\frac{h}{2}\right) + 2^p 4^p D\left(\frac{h}{4}\right) \right)}{\left(2^p (2 4^p + 1) - 3 4^p \right) h^p}, C_2 = \frac{(2^p - 4^p) h D(h) + \left(2^p (4^p - 1) D\left(\frac{h}{2}\right) - 2^p 4^p D\left(\frac{h}{4}\right) + 4^p D\left(\frac{h}{4}\right) \right) h}{2^p (2 4^p + 1) - 3 4^p} \right]$$

```
(%i11) /* Compute the absolute error committed for D(h) */
err1: abs(ratsimp(subst(C1C2, q1) - D(h)));
```

err1
$$\frac{\left| \left((2^{p+1} + 1) 4^p - 2^p \right) D(h) + (2^p - 2^{p+2} 4^p) D\left(\frac{h}{2}\right) + (2^{p+1} - 1) 4^p D\left(\frac{h}{4}\right) \right|}{\left| (2^{p+1} - 3) 4^p + 2^p \right|}$$

```
(%i13) factor(subst(p = 1, err1)); /* Forward or backward difference method p = 1 */
factor(subst(p = 2, err1)); /* Central difference method p = 2 */
```

(%o12) $\left| 3 D(h) - 5 D\left(\frac{h}{2}\right) + 2 D\left(\frac{h}{4}\right) \right|$

(%o13) $\left| 5 D(h) - 9 D\left(\frac{h}{2}\right) + 4 D\left(\frac{h}{4}\right) \right|$